A Two-Point Obstruction to the Literal Finite-Cardinality Question in arXiv:2402.03189

agent_lane_16

2026-06-15

Source and Question

The source is Jan Bima, *Nagata Dimension and Lipschitz Extensions Into Quasi-Banach Spaces*, arXiv:2402.03189v1. Definition 2 defines the p-trace $\mathfrak{t}_p(\mathcal{N}, \mathcal{M})$ and the absolute p-extendability constant

$$\mathfrak{ae}_p(\mathcal{N}) = \sup_{\mathcal{M} \supseteq \mathcal{N}} \mathfrak{t}_p(\mathcal{N}, \mathcal{M}).$$

On page 2 the source notes that $\mathfrak{ae}_p(\mathcal{N}) = 1$ for every two-point metric space $\mathcal{N}$ and $0 < p < 1$.

definition.

Definition 2. For a metric space $\mathcal{N}$ and each $0 < p \le 1$, we define the *p-trace* $\mathfrak{t}_p(\mathcal{N}, \mathcal{M})$ *of $\mathcal{N}$ in $\mathcal{M} \supseteq \mathcal{N}$* to be the infimum over all $C \in (0, \infty]$ such that for any p-Banach space Z, any Lipschitz map $f : \mathcal{N} \to Z$ has a Lipschitz extension $\widetilde{f} : \mathcal{M} \to Z$ with $\operatorname{Lip} \widetilde{f} \le C \operatorname{Lip} f$.

We define the *absolute p-extandability constant* $\mathfrak{ae}_p(\mathcal{N}) = \sup\{\mathfrak{t}_p(\mathcal{N}, \mathcal{M}) : \mathcal{M} \supseteq \mathcal{N}\}$. If $\mathfrak{ae}_p(\mathcal{N}) < \infty$, we say that $\mathcal{N}$ is *absolutely p-extendable.*

To better motivate the notions we just introduced, particularly in relation to the p-Banach setting where $0 < p \le 1$, we can consider extensions from finite subsets. Note that it is easy see that finite metric spaces are absolutely p-extendable. However, determining the lower and upper estimates on the extendability constant $\mathfrak{ae}_1(n) = \sup\{\mathfrak{ae}_1(\mathcal{N}) : |\mathcal{N}| \le n\}$ is a significant open problem, see Lee and Naor [34] and Naor and Rabani [38]. In general, the problem becomes even more challenging for $0 < p < 1$. While it can easily be observed that $\mathfrak{ae}_p(\mathcal{N}) = 1$ for any two-point metric space $\mathcal{N}$ and $0 < p < 1$, the absolute extendability constant typically increases as p approaches zero. In particular, we have the following result, which is intriguing when

Question 33, on page 20, asks whether

$$\lim_{p \to 0} \mathfrak{ae}_p(n) = \infty$$

for each $n \ge 2$. It further asks whether, for every $n, m \in \mathbb{N}$,

$$\sup\{\lim_{p \to 0} \mathfrak{t}_p(\mathcal{N}, \mathcal{M}) : \mathcal{N} \subset \mathcal{M},\ |\mathcal{N}| \le n,\ |\mathcal{M} \setminus \mathcal{N}| \le m\} = m + 1.$$

lished a basic counterexample demonstrating that the p-trace typically does indeed depend on p. Consequently, we are curious to see how this behavior is exhibited across the different classes of spaces that we have considered in this paper, and whether we can derive lower estimates on this.

In the following discussion, whenever $\mathcal{N}$ is a metric space with $\mathfrak{ae}_1(\mathcal{N}) < \infty$, we denote $\mathfrak{q}(\mathcal{N}, p) = \mathfrak{ae}_p(\mathcal{N})/\mathfrak{ae}_1(\mathcal{N})$ for each $0 < p \leq 1$. It is easy to see that $\lim_{p\to 0} \mathfrak{q}(\mathcal{N}, p)$ exists as $p \mapsto \mathfrak{ae}_p(\mathcal{N})$ does not increase in p.

Question 32. Is it true that for each $n \geq 2$, we have $\sup\{\lim_{p\to 0} \mathfrak{q}(\mathcal{N}, p) : \mathcal{N}$ is a doubling with $\lambda_{\mathcal{N}} \leq n\} = \infty$? More specifically, can it be shown that for each fixed $n \geq 2$, $p \mapsto \log \sup\{\mathfrak{q}(\mathcal{N}, p) : \lambda_{\mathcal{N}} \leq n\}$ grows proportionally to $1/p$ as $p \to 0$? What about if we consider $\mathcal{N}$ to only have a finite Nagata dimension at most d with constant γ?

In relation to the example of Theorem 30, we would like to know whether, if the answer to the very first question in Question 32 is positive, it could potentially be identified within examples involving finite metric spaces.

Recall that by Corollary 16, we have $\mathfrak{ae}_p(n) < \infty$ for every $n \in \mathbb{N}$ and $0 < p \leq 1$ (we have a trivial estimate $\lambda_{\mathcal{N}} \leq n$ for every n-point metric space $\mathcal{N}$). Interestingly, observe also that in the proof of Theorem 30, while the optimal projection was identified as a weighted sum of two points in the target Lipschitz free p-space $\mathcal{F}_p(\mathcal{N})$ for $p = 1$, the norm of that particular extension would become excessively large as $p \to 0$. In particular, as $p \to 0$, the Lipschitz norm of the optimal extension converged to that of a "trivial" extension that projected the additional point to a single point in $\mathcal{F}_p(\mathcal{N})$. It is clear that such a projection does not require the underlying linear structure of the target space.

At the same time, note that a remark to [11, Theorem 1.1] shows that for each $m \in \mathbb{N}$, there exist spaces $\mathcal{N} \subset \mathcal{M}$ with $|\mathcal{N}| = 2$ and $|\mathcal{M} \setminus \mathcal{N}| = m$, such that if f is the identity map $f : \mathcal{N} \to \mathcal{N}$, then $\operatorname{Lip} f' \geq (m+1) \operatorname{Lip} f$ for any extension $f' : \mathcal{M} \to \mathcal{N}$ of f. Consequently, we pose the following question.

Question 33. Is it true that $\lim_{p\to 0} \mathfrak{ae}_p(n) = \infty$ for each $n \geq 2$? Additionally, for any n, $m \in \mathbb{N}$, is it true that $\sup\{\lim_{p\to 0} \mathfrak{t}_p(\mathcal{N}, \mathcal{M}) : \mathcal{N} \subset \mathcal{M}$ with $|\mathcal{N}| \leq n$ and $|\mathcal{M} \setminus \mathcal{N}| \leq m\} = m+1$? What if we relax the condition that $|\mathcal{N}| \leq n$?

Remark 34. We wish to acknowledge that a generalization of the extension theorem from doubling spaces to spaces with finite Nagata dimension has been recently independently discovered by Basso [10], in the narrower context of the Banach setting with $p = 1$. This is the content of Theorem II and, specifically, Theorem 14. Interestingly, both methods of proof share similari-

Idea of the Proof

The endpoint $n = 2$ is rigid for a simple reason: a Lipschitz map from two points into any p-Banach space has only one affine direction. A scalar McShane extension of the coordinate along that direction gives a constant-preserving extension to every ambient metric space. Therefore the two-point absolute constant never grows as $p \to 0$.

Theorem

Let $0 < p \le 1$. If $\mathcal{N}$ is a two-point metric space, then $\mathfrak{ae}_p(\mathcal{N}) = 1$; one-point spaces have extension constant no larger than this. Consequently,

$$\lim_{p \to 0} \mathfrak{ae}_p(2) = 1,$$

so the first assertion of Question 33 is false as literally stated. Moreover, for every $m \ge 1$,

$$\sup\{\lim_{p \to 0} \mathfrak{t}_p(\mathcal{N}, \mathcal{M}) : \mathcal{N} \subset \mathcal{M},\ |\mathcal{N}| \le 2,\ |\mathcal{M} \setminus \mathcal{N}| \le m\} = 1,$$

not $m + 1$.

Proof. It is enough to handle the two-point case, so let $\mathcal{N} = \{u, v\}$ and put $D = d(u, v) > 0$. Let $\mathcal{M} \supseteq \mathcal{N}$ be any metric space, let Z be any p-Banach space, and let $f : \mathcal{N} \to Z$ be Lipschitz.

Choose a 1-Lipschitz scalar extension $\alpha : \mathcal{M} \to [0, D]$ of the data $\alpha(u) = 0$, $\alpha(v) = D$; for example

$$\alpha(x) = \min\{D, d(x, u)\}.$$

Define

$$F(x) = f(u) + \frac{\alpha(x)}{D}\big(f(v) - f(u)\big) \qquad (x \in \mathcal{M}).$$

Then $F(u) = f(u)$ and $F(v) = f(v)$. For $x, y \in \mathcal{M}$, homogeneity of the quasi-norm gives

$$\|F(x) - F(y)\| = \frac{|\alpha(x) - \alpha(y)|}{D}\,\|f(v) - f(u)\| \le \operatorname{Lip}(f)\,|\alpha(x) - \alpha(y)| \le \operatorname{Lip}(f)\,d(x, y).$$

Thus every f extends with no loss of Lipschitz constant, and $\mathfrak{t}_p(\mathcal{N}, \mathcal{M}) \le 1$. The reverse inequality is forced by any nonconstant map on $\mathcal{N}$, so $\mathfrak{t}_p(\mathcal{N}, \mathcal{M}) = 1$ whenever $|\mathcal{N}| = 2$. Taking the supremum over all $\mathcal{M} \supseteq \mathcal{N}$ yields $\mathfrak{ae}_p(\mathcal{N}) = 1$.

It follows that $\mathfrak{ae}_p(2) = \sup_{|\mathcal{N}| \le 2} \mathfrak{ae}_p(\mathcal{N}) = 1$ for all $0 < p \le 1$, and hence its limit as $p \to 0$ is 1. The displayed finite-trace supremum with $n = 2$ is also 1: two-point domains always have $\mathfrak{t}_p(\mathcal{N}, \mathcal{M}) = 1$, one-point domains have trace constant no larger than 1, and the value 1 is attained by taking any two-point domain. □

Limitations

This packet only addresses the literal $n = 2$ edge case in Question 33. It does not settle the intended finite problem for $n \ge 3$, nor the doubling or finite-Nagata-dimension growth question in Question 32.

Novelty and Verification Notes

A bounded novelty search was run for the phrases `ae_p(2)`, `absolute p-extendability two-point`, `two-point metric space absolute p-extendable`, and $\mathfrak{ae}_p(n)$. No separate literature answer was found. The core two-point fact is already explicitly observed on page 2 of the source paper; the contribution here is the quantifier check against the later formulation of Question 33.

Human Review Recommendation

Review as an erratum-style counterexample to the literal statement. If the intended reading is silently restricted to $n \geq 3$, this packet should be classified as a scope correction rather than a substantive solution of the main open problem.

References

Jan Bima, *Nagata Dimension and Lipschitz Extensions Into Quasi-Banach Spaces*, arXiv:2402.03189v1, 2024.